

5-6-1: Examining Network Storage

At the end of this episode, I will be able to:

1. Identify and explain the purpose of network storage technologies, given a scenario.

Learner Objective: *Identify and explain the purpose of network storage technologies, given a scenario.*

Description: In this episode, the learner will examine various network storage technologies such as Storage Area Networks (SANs) and Network-attached Storage (NAS). We will explore technologies such as Fibre Channel, connections and network file systems.

- What is are Storage Area Networks?
 - o are high-speed, dedicated networks that provide access to consolidated, block-level data storage.
 - o Designed to handle large volumes of data in enterprise environments and improved performance
 - o Connections protocols
 - ♣ Fibre Channel (FC)
 - ♣ Fibre Channel over Ethernet (FCoE)
 - ♣ InfiniBand
 - ♣ iSCSI
- What is Fibre Channel?
 - o Fibre Channel Overview
 - o A high-speed network technology widely used in storage networking.
 - o Support multiple transmission speeds
- What are some of the Fibre Channel Speeds?
 - o 1 Gbps (1GFC) - The initial speed, foundational for early SANs.
 - o 2 Gbps (2GFC) - Doubled the data transfer rate, improving performance.
 - o 4 Gbps (4GFC) - Enhanced speed for increased data demands.
 - o 8 Gbps (8GFC) - Further doubled speeds to support growing storage needs.
 - o 16 Gbps (16GFC) - Introduced to manage larger, more data-intensive environments.
 - o 32 Gbps (32GFC) and 128 Gbps (128GFC) - Latest standards, providing high-speed connections for modern data centers.
- Do these speeds translate to versions?
 - o Each speed increment often corresponds with a new version of the Fibre Channel standard, incorporating improvements in technology, security, and efficiency.
- What are the connection technologies or protocol used?:
 - o SMB (Server Message Block)
 - ♣ Enables network file sharing
 - ♣ SMB 1.0 for legacy system
 - ♣ SMB 2.0 for improved performance and security
 - ♣ SMB 3.0 for enhanced capabilities and encryption.
 - o NFS (Network File System)
 - ♣ Facilitates file access over networks
 - ♣ NFSv3 for stateless protocol operations
 - ♣ NFSv4 adding stateful operations and improved security
 - ♣ NFSv4.1 introducing parallel NFS (pNFS) for performance.
 - o iSCSI (Internet Small Computer Systems Interface)
 - ♣ Allows for the transmission of block-level data over IP networks
 - ♣

- It utilizes iSCSI initiators to send data requests and iSCSI targets to receive and store them, streamlining storage management and access.

o Fibre Channel over Ethernet (FCoE)

- ♠ Encapsulates Fibre Channel frames over Ethernet network

- ♠ Enables the convergence of storage and IP protocols over a single network infrastructure.

- ♠ InfiniBand - A high-performance networking technology that offers very low latency and high throughput

- ♠ Used in high-performance computing and enterprise data centers.

o FTP (File Transfer Protocol)

- ♠ A standard network protocol used for the transfer of files between a client and server on a computer network-

o SFTP (Secure File Transfer Protocol)

- ♠ Enhances FTP by providing secure file transfer capabilities through the use of SSH encryption.